

Junhyeok Kil

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Research interests

Optimization in stochastic networks with higher-order interactions, with applications in business, engineering, and AI

Education

- 2026–present PhD in Quantitative Methods, Purdue University
Advisor: Tongseok Lim
- 2020–2025 BS in Data Science, Mathematics, and Statistics, Purdue University

Grants, honors & awards

- 2026–2027 Google TPU Research Award, \$25,000 in compute credits (PI: Tongseok Lim)
- 2024–2025 Gordon L. Walker Scholarship in Mathematics, \$3,000, Purdue University
- 2024 Finalist, Munich Re Cup, New York (\$400 stipend, full travel & lodging)
- 2024 Undergraduate Quantum Summer School, Northwestern University (full travel & lodging)

Publications

- 2025 **Junhyeok Kil**, Akshay Jacob Thomas, Eduardo Barocio, and Ilias Bilionis
“Physics-informed neural networks to accelerate heat transfer predictions in additive manufacturing”
SAMPE Conference Proceedings, pp. 1117–1130 [[Paper](#)]

Talks

- 2026 *Physics-informed neural networks for 3D heat transfer prediction in additive manufacturing*, Purdue AI Research Showcase [[Poster](#)]
- 2025 *Physics-informed neural networks for 3D heat transfer predictions in additive manufacturing*, SAMPE Conference and Exhibition, Indianapolis [[Slides](#)]
- 2024 *Exploring graph connectivity through the adjacency matrix in spectral graph theory*, Directed Reading Program, Purdue Department of Mathematics (advisor: Anurag Sahay) [[Slides](#)]
- 2023 *Predicting shape change compensation of composite printed tooling*, Purdue Fall Expo [[Poster](#)]

Teaching

Instructor

Fall 2026 QM 30500: Business Statistics, Purdue University

Teaching assistant

Fall 2025 CS 47100: Introduction to Artificial Intelligence, Purdue University

Fall 2024 CS 24200: Introduction to Data Science, Purdue University

Spring 2024 CS 17600: Data Engineering in Python, Purdue University